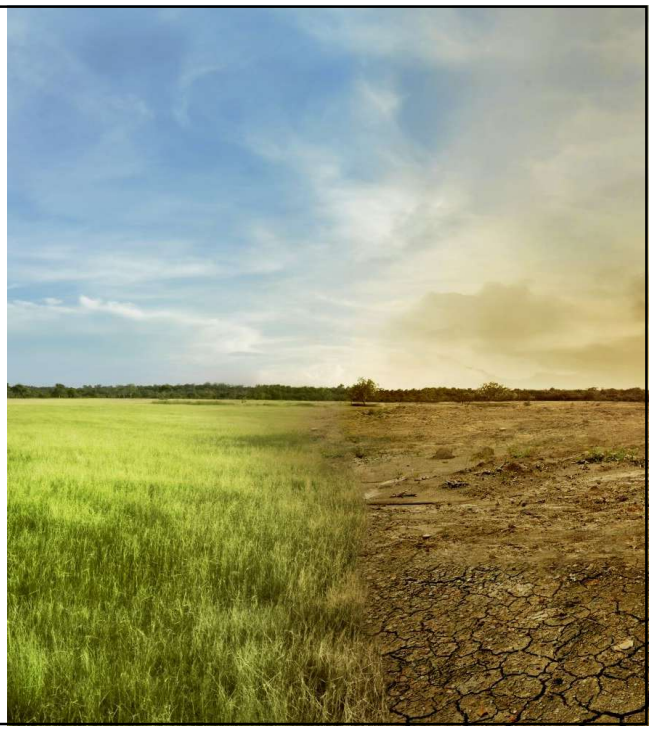
Soil Pollution

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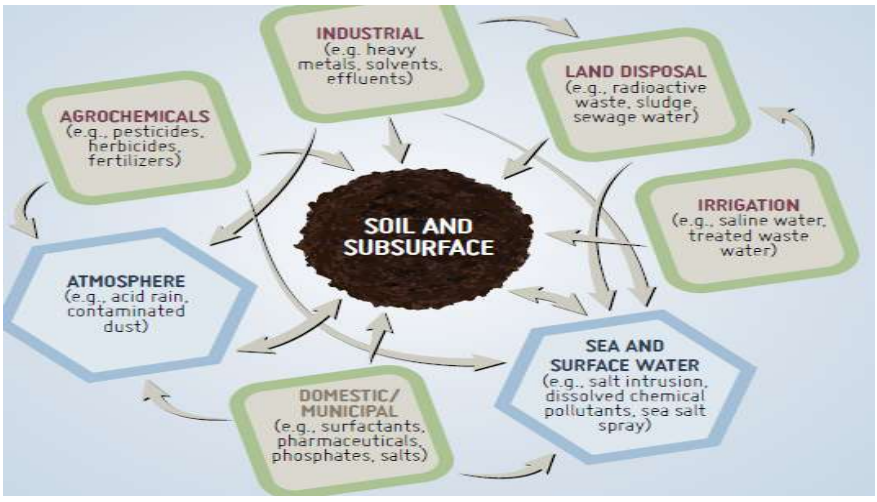
Learning Objectives

After taking this class, the students will be able to:

- *understand* what soil pollution is
- acknowledge the urgency of dealing with soil pollution
- acknowledge the practices to mitigate soil pollution

What is soil pollution?

Soil Pollution



Soil Pollution

- The presence of toxic chemicals (pollutants or contaminants) in soil, in **high enough concentrations** to pose a risk to human health and/or the ecosystem.
- Sources of Soil Pollution
  - Irrigation with polluted water
  - Pollution from solid waste disposal
  - Pesticide and fertilizer
  - Pollutants from waste gas, especially the particulates
  - Mining

Some Facts of Soil and Soil Pollution

- Soils are the second largest active store of carbon, and crucial in fighting the climate crisis.
- Unsustainable agriculture is one of the main causes of soil pollution. Some 109 million tons of synthetic nitrogen fertilizers were applied worldwide in 2018. The excess nitrogen alters biological cycles in soil and is released into the atmosphere in the form of N_2O, resulting in emissions of 700,000 CO_2 equivalents. In 2017, agriculture emissions accounted for 20 percents of all human activities.
- Industrial pollution, mining, farming and poor waste management are poisoning soils, with the "polluter pays" principle absent in many countries. Since the Industrial Revolution, about 135bn tons of soil has been lost from farmland and, given that it takes thousands of years for soils to form, urgent protection and restoration of the soils that remain is needed.

Regional Status and Trends of Soil Pollution:

Asia and the Pacific

- The risks of soil pollution in industrial regions in China are more severe than in agricultural regions.
- In general, trace element pollution is more serious in southeast China than in northwest China.
- The southern provinces and Liaoning province should be prioritized for pollution control.
- Roadside soil pollution in China's urban and suburban areas is caused by the deposition of toxic chemicals emitted by vehicles.



Hierarchical chart showing the main sources of soil pollution

Regional Status and Trends of Soil Pollution:

Eastern Europe, Caucasus and Central Asia (Eurasia)

- There **are no regional or comprehensive national inventories** of polluted sites available for the region.
- 424 polluted sites in eight countries of the Eurasian region (Armenia, Azerbaijan, Kazakhstan, Kyrgyzstan, the Russian Federation, Tajikistan, Ukraine and Uzbekistan) are reported.
- Trace element pollution occurs at 54 percent of the sites, with lead (16 percent), and arsenic (10 percent), together with radionuclides including uranium (15 percent)



Hierarchical chart showing the main sources of soil pollution

Regional Status and Trends of Soil Pollution:

Europe

- **37 of the 38 countries release regular reports and indicators** regarding the state of soils and the environment;
- In EU member states, 2.8 million sites are suspected to be potentially polluted; only one quarter are included in national registries
- In Europe, there are about 650,000 sites identified as potentially polluted, and that are inserted in national and/or regional inventories.
- Industrial and commercial activities together with waste disposal and treatment, have caused almost two thirds of the point source soil pollution.



Hierarchical chart showing the main sources of soil pollution

Regional Status and Trends of Soil Pollution:

Latin America and the Caribbean



- This region has the world's largest reserves of arable land. During the last 50 years the region's agricultural surface area increased from 561 million ha to 741 million ha.
- The main agricultural practices contributing to soil pollution in the region are the mismanagement of mineral fertilizers and pesticides.



Hierarchical chart showing the main sources of soil pollution

Regional Status and Trends of Soil Pollution:

Near East and North Africa

- In the NENA region, solid waste management mainly consists of landfilling or uncontrolled open dumps.
- The landfills and dumps often receive domestic, hospital and industrial wastes, which are often burned.



Hierarchical chart showing the main sources of soil pollution

Regional Status and Trends of Soil Pollution:

North America

- The environment protection agencies of both countries receive an annual report from the industries on the amount of toxic substances released in the three environmental compartments: water, air and land.
- The dataset is publicly available in the form of inventories called the **National Pollutant Release Inventory (NPRI) for Canada and the Toxics Release Inventory (TRI) for the US**.
- There are thousands of polluted sites in both countries, of varying size and significance.
- In 2017, US EPA and its state partners reported monitoring approximately 640,000 to 1,319,100 facilities to prevent releases to communities.
- In July 2019, there were 23,663 federal sites included in Canada's Federal Inventory of Contaminated Sites, with 16,845 sites listed as closed because remediation is not feasible.



Hierarchical chart showing the main sources of soil pollution

Regional Status and Trends of Soil Pollution:

Sub-Saharan Africa

- This region has significant mineral resources and oil deposits.
- In 2012, approximately 81 million tonnes of Municipal Solid Waste (MSW) was generated in sub-Saharan Africa. It is projected that by 2025, the volume of MSW will grow to 244 million tonnes.



Hierarchical chart showing the main sources of soil pollution

Some Facts of Soil and Soil Pollution

- According to the Chinese MEP, 16% of all Chinese soils and 19% of its agricultural soils are categorized as polluted (CCICED, 2015).
- There are also approximately 3 million potentially polluted sites in the European Economic Area and cooperating countries in the West Balkans (EEA, 2014) and more than 1300 polluted or contaminated sites in the US are included on the Superfund National Priorities List (US EPA, 2013)
- The total number of contaminated sites is estimated at 80,000 across Australia (DECA, 2010).

Investigation on Soil Pollution in China

- Investigation period：2005.4 – 2013.12
- Area: Mainland China, 63 million km^2

- The national total soil contamination exceedance rate is 16.1%, with the proportions of slight, mild, moderate, and severe contamination points being 11.2%, 2.3%, 1.5%, and 1.1%, respectively.

- The types of pollution are mainly inorganic, followed by organic, with a smaller proportion of compound pollution. The number of exceedance points for inorganic pollutants accounts for 82.8% of all exceedance points.

- Soil contamination in the south is more severe than in the north. Areas such as the Yangtze River Delta, the Pearl River Delta, and the old industrial bases in the Northeast have more prominent soil pollution issues. The southwest and central-south regions have a wider range of soil heavy metal exceedances.

Investigation on Soil Pollution in China

Land Type	All Kinds of Pollution	Slightly Polluted	Mildly Polluted	Medium Polluted	Heavily Polluted
Arable Land	19.4%	13.7%	2.8%	1.8%	1.1%
Woodland	10%	5.9%	1.6%	1.2%	1.3%
Grassland	10.4%	7.6%	1.2%	0.9%	0.7%
Unused Land	11.4%	8.4%	1.1%	0.9%	1.0%

Investigation on Soil Pollution in China



Reflective Question:

How is soil pollution different from air pollution, water pollution, or solid waste pollution?

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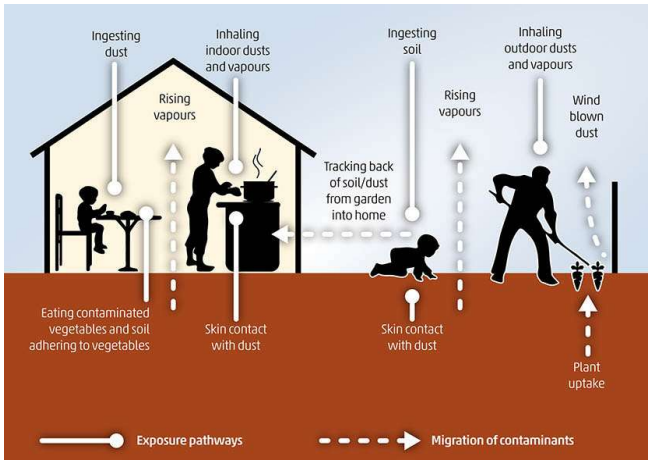
- Hidden and delayed impact
- Accumulative and less diffuse
- Unevenly distributed across different location
- Irreversible or hard to reverse
- Difficult to control

What are the impacts of soil pollution?

Impacts on Human Health

- Short term effects:
  - Headaches, nausea, and vomiting
  - Coughing, pain in the chest, and wheezing
  - Irritation of the skin and the eyes
  - Fatigue and weakness

- Long term effects:
  - Permanent damage to the nervous system.
  - Depression of the CNS (Central Nervous System)
  - Damage to vital organs such as the kidney and the liver
  - Higher risk of developing cancer.

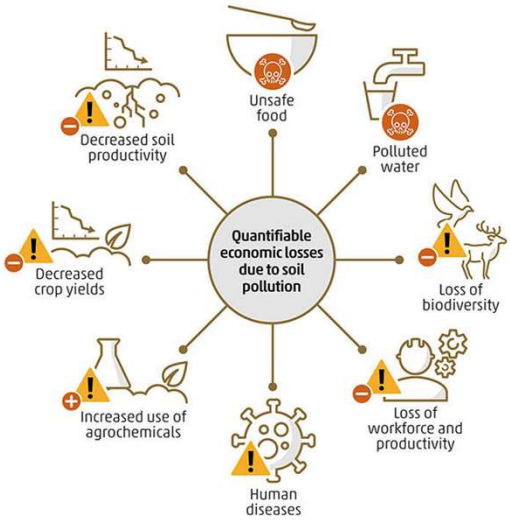


Impacts on Plants and Animals

- a decrease in the availability of nutrients, inhospitable for the growth of plant life

- Soils contaminated with inorganic aluminum can prove toxic to plants.

- loss/extinction of many desirable animal species



Impacts on Ecosystem

- Soil pollution can be a direct contributor to air and water pollution.
- It can also contribute to acid rain (by releasing huge quantities of ammonia into the atmosphere).
- Poor soil quality and texture. Crop yield is greatly affected. In China, over 12 million tons of grain (worth approximately 2.6 billion USD) is found to be unfit for human consumption due to contamination with heavy metals.



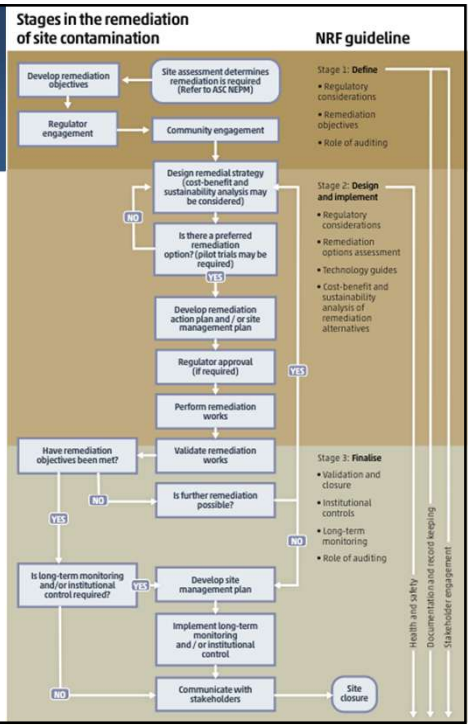
How do we manage soil pollution?

Management of Soil Contamination and Pollution

- **Identification and risk assessment**
  - identify sites where pollution may exist (e.g., historical analysis, knowledge of accident sites)
  - whether they are posing any risk to the environment and to human health
  - The determination of the degree of pollution and its potential for harm is carried out in a risk assessment framework.
- **Management of contaminated site**
  - Adaptation to the presence of contaminants and follows the general principles of Sustainable Soil Management (SSM);
  - Change in land use (e.g. a change to crops that do not bio-accumulate and biomagnify the particular contaminant that has been detected)
  - Reduce or eliminate dispersal of contaminants by wind or water erosion by implementing soil conservation measures

Management of Soil Contamination and Pollution

- **Management of polluted site**
  - Soil remediation (reducing the concentration of the contaminant in the soil) is a permanent approach to managing pollution.
  - Remediation of polluted sites is a site-specific approach that includes identification, characterization, risk assessment and selection of remediation technologies, and is mainly focused on point-source pollution



Policy Framework of the Chinese Soil Management System

- Soil Pollution Prevention and Control Law (土壤污染防治法)
- Action Plan on Prevention and Control of Soil Pollution (土壤污染防治行动计划)
- Measures for Soil Environmental Management of Agricultural Land/Contaminated Land/Industrial Land
- Soil Environmental Quality Risk Control Standard for Soil Contamination of Agricultural Land/Development Land